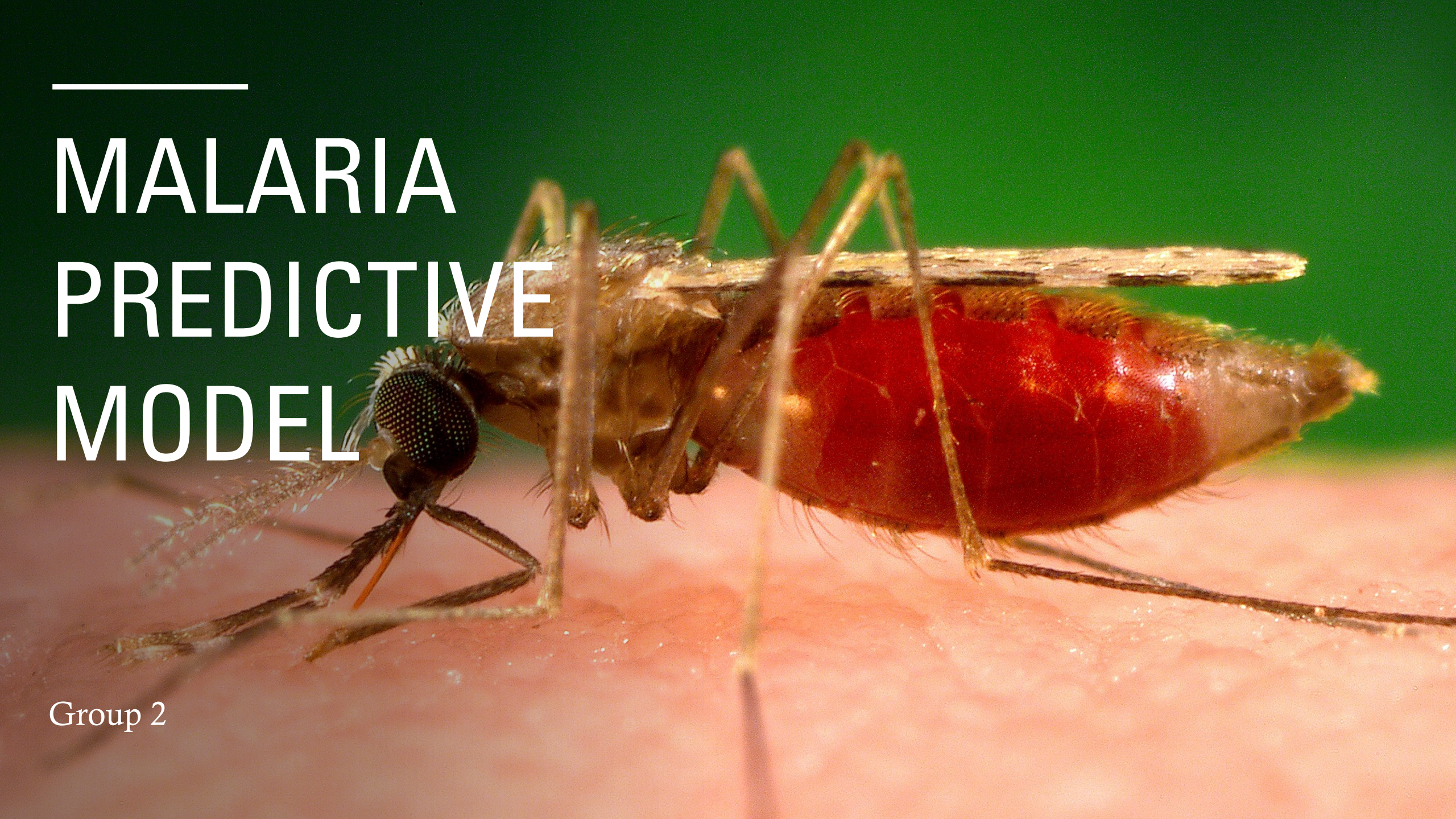

MALARIA PREDICTIVE MODEL

Group 2





BUSINESS UNDERSTANDING

Malaria remains a major public health challenge in Kenya, with 10 counties accounting for ~95% of the national burden—Turkana among the highest.

Kakuma Ward alone manages over 65,000 malaria cases annually, with increasingly unpredictable seasonal peaks due to climate variability.

Reactive responses in ASAL regions lead to stock-outs, Health Care Worker burnout, and excess morbidity, especially among pregnant women and children and mortality.

This capstone proposes a **climate-informed malaria early warning and response platform**, using weather data and historical trends to forecast outbreaks and guide timely interventions.

PROBLEM STATEMENT AND STAKEHOLDERS

- **How can high-resolution temporal weather data (rainfall, temperature, windspeed, humidity) combined with historical malaria case trends improve the accuracy of short-term malaria outbreak forecasts in endemic regions?**
 - Stakeholders:
 - Ministry of health
 - Facility health management teams
 - Community health promoters
 - Data scientist.
-



DATA UNDERSTANDING

- **Data Acquisition:** We used publicly available data from **District Health Information System (DHIS 2)** and **Open-Meteo**
 - **Malaria Case Data**

The data was extracted from facility-level records in **DHIS2**, including case counts and malaria positivity rates for Kakuma Ward.
 - **Weather Data**

Pulled from **Open-Meteo APIs**, covering key variables:

 - Rainfall
 - Temperature
 - Humidity
 - Wind speed
 - Soil moisture
-



DATA PREPARATION

Preprocessing Workflow

Temporal alignment by week/location; impute rare gaps; apply log1p transform; and residuals; engineer lags and rolling statistics.

Cleaning Challenges & Solutions

Handle reporting backfills, outliers, heteroscedasticity, and multicollinearity, multiplicative and additive decomposition, regularization,

Minimum Dataset Size

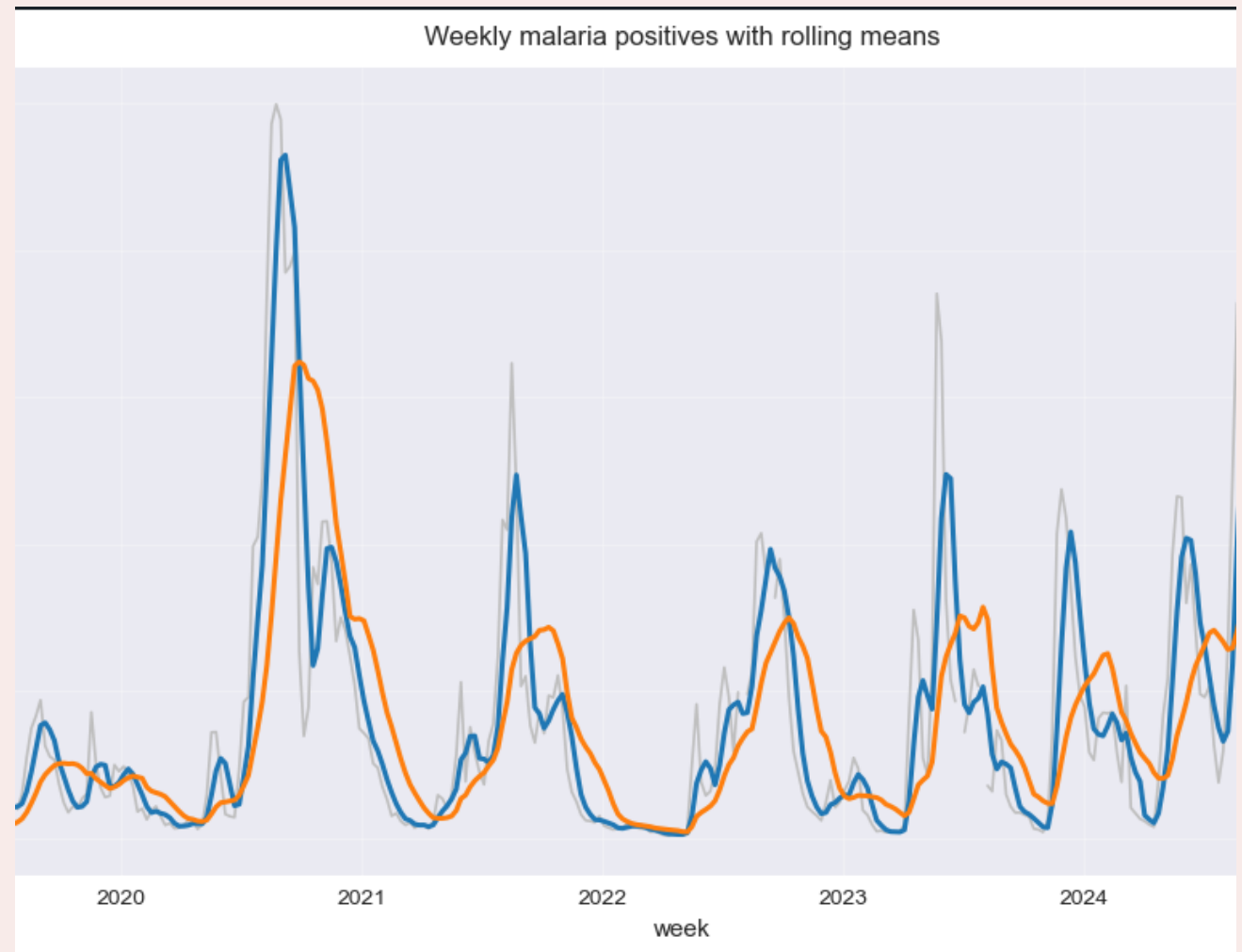
At least **300 weekly observations** post-cleaning.

Visualization Techniques

Rolling means, seasonal profiles (mean $\pm$ SD), anomaly heatmaps (z-scores), ACF/PACF plots, lagged correlations, and STL decomposition panels.

MALARIA TRENDS

- Weekly malaria cases show **seasonal peaks**, with major spikes in **2020 and 2023**.
- **Rolling averages (4-week & 12-week)** reveal short-term fluctuations and a rising long-term trend.
- **Climate variability** has made peak timing less predictable, complicating response planning.
- Trend data supports **seasonality-aware forecasting** and **early intervention strategies**.



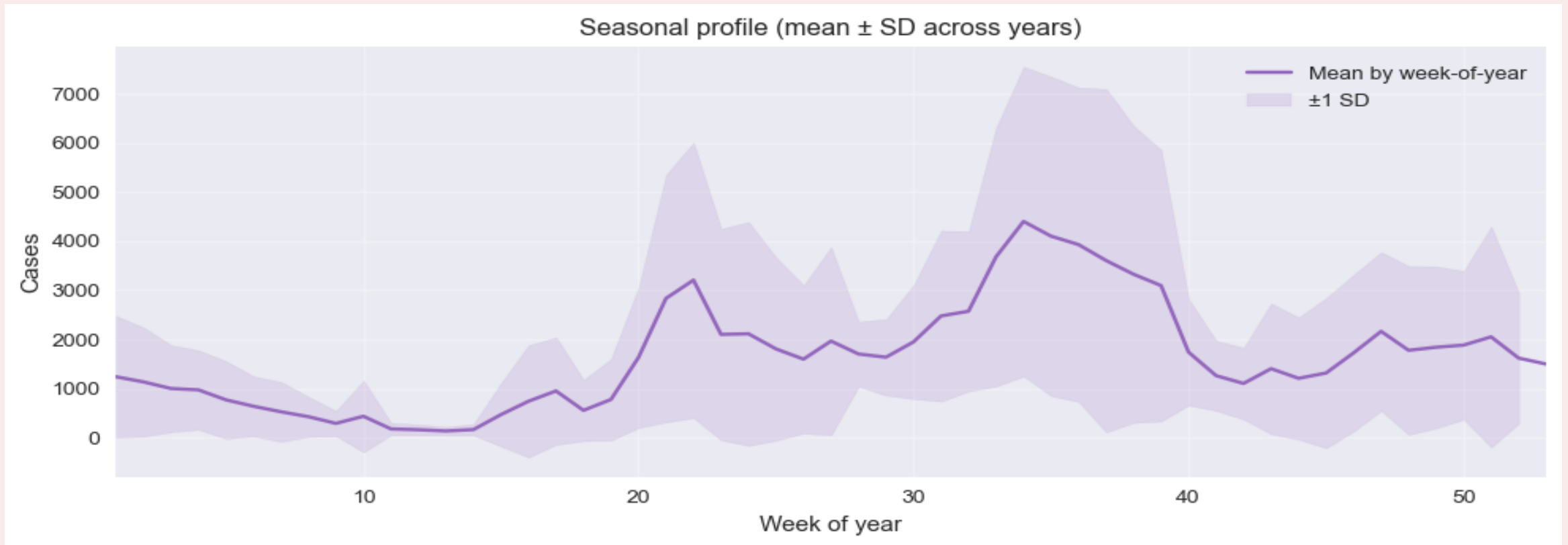
SEASONAL PROFILE GRAPH



Two major peaks: around **week 21 (May)** and **week 33 (August)**.

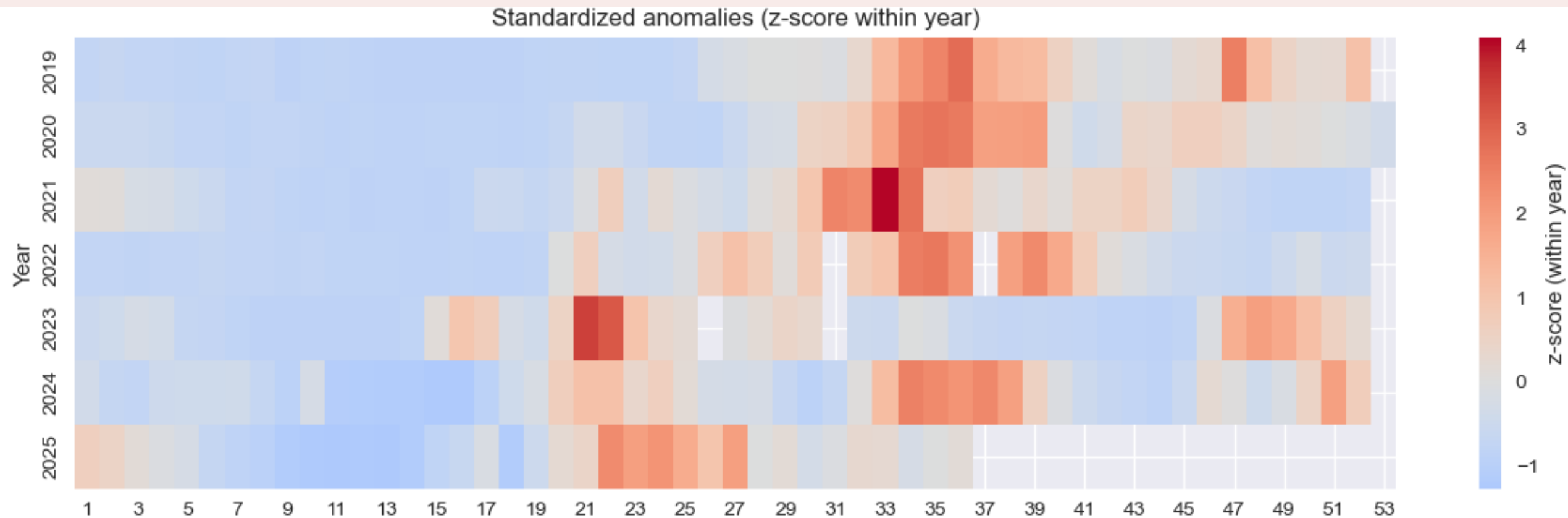
Shaded band shows variability—some years have much higher spikes than others.

Useful for **timing interventions** (e.g., Indoor Residual Spraying, Long Lasting Insecticide Nets distribution) and modeling **seasonality features**.



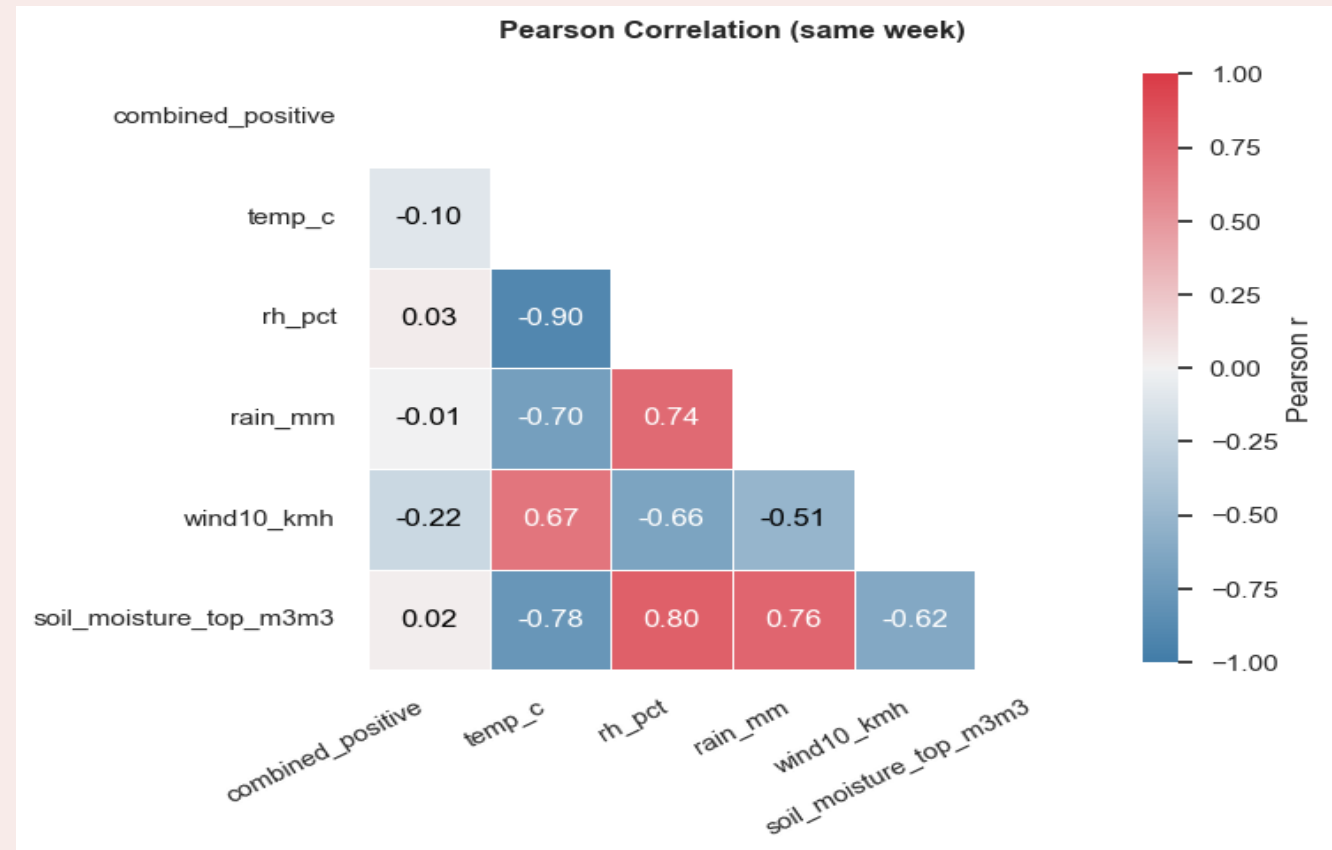
ANOMALY HEATMAP

- **Red zones** indicate weeks with significantly higher-than-normal cases (e.g., outbreaks).
- **Blue zones** show lower-than-normal activity.
- Consistent anomalies around **weeks 21 and 33**, confirming seasonal peaks.
- Intense red clusters in **2020 and 2023** align with known outbreak years.



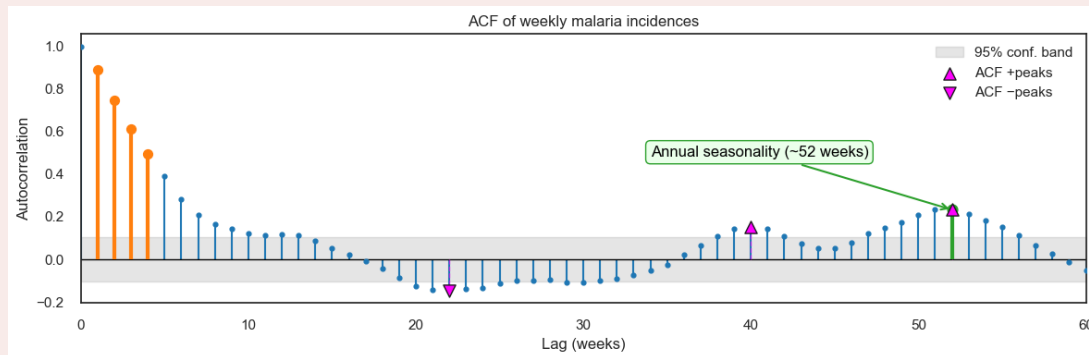
CORRELATION ANALYSIS

- **Lagged weather effects are key:** Malaria cases correlate more strongly with weather conditions from 4–5 weeks earlier than with current conditions.
- **Rainfall, humidity, and soil moisture** show positive correlations at 4–5 week lags, supporting their role in mosquito breeding and transmission risk.
- **Temperature and wind speed** show negative correlations at similar lags, suggesting cooler, calmer conditions may favor malaria spread.

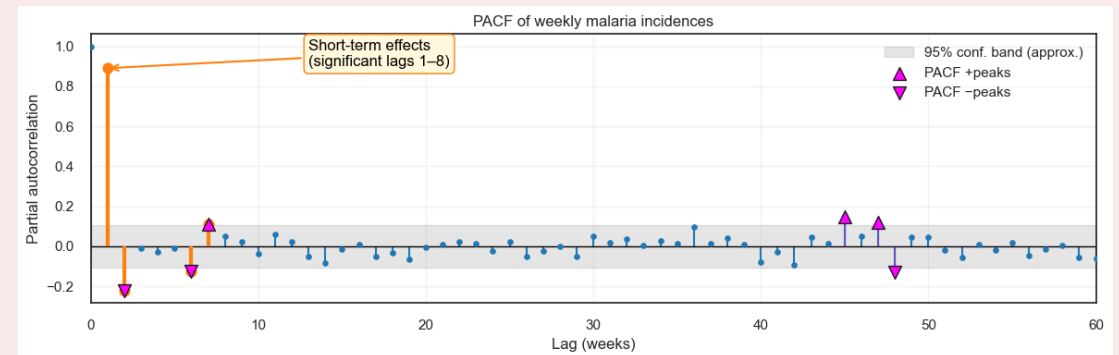


AUTOCORRELATION

- **Strong short-term dependence:** Malaria cases show high autocorrelation at lags 1–4 weeks, especially at lag 1 ($r \approx 0.89$), indicating persistence in weekly transmission.
- **Annual seasonality detected:** A significant peak at lag 52 confirms yearly cycles, likely driven by seasonal weather patterns.
- **Mid-year dip pattern:** Negative autocorrelations around lag 22 suggest a temporary decline in cases before seasonal peaks resume.



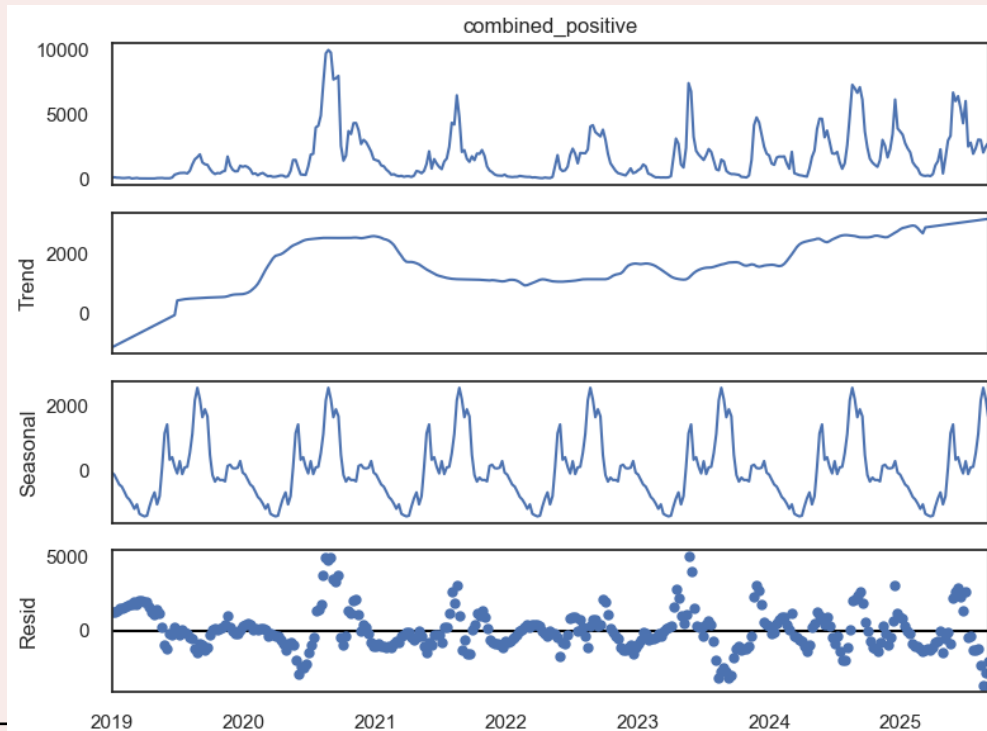
- **Strong short-term autoregression:** Dominated by a large spike at lag 1 (≈ 0.89) and notable spikes at lags 2, 6, and 7.
- **No direct seasonal AR effect:** Most coefficients beyond lag 8 fall within the 95% confidence band; weak signals near lags 45–47.
- **Model implication:** Suggests a low-order autoregressive component, suitable for models like **ARIMA with $p \approx 1-2$** .



SEASONAL DECOMPOSITION

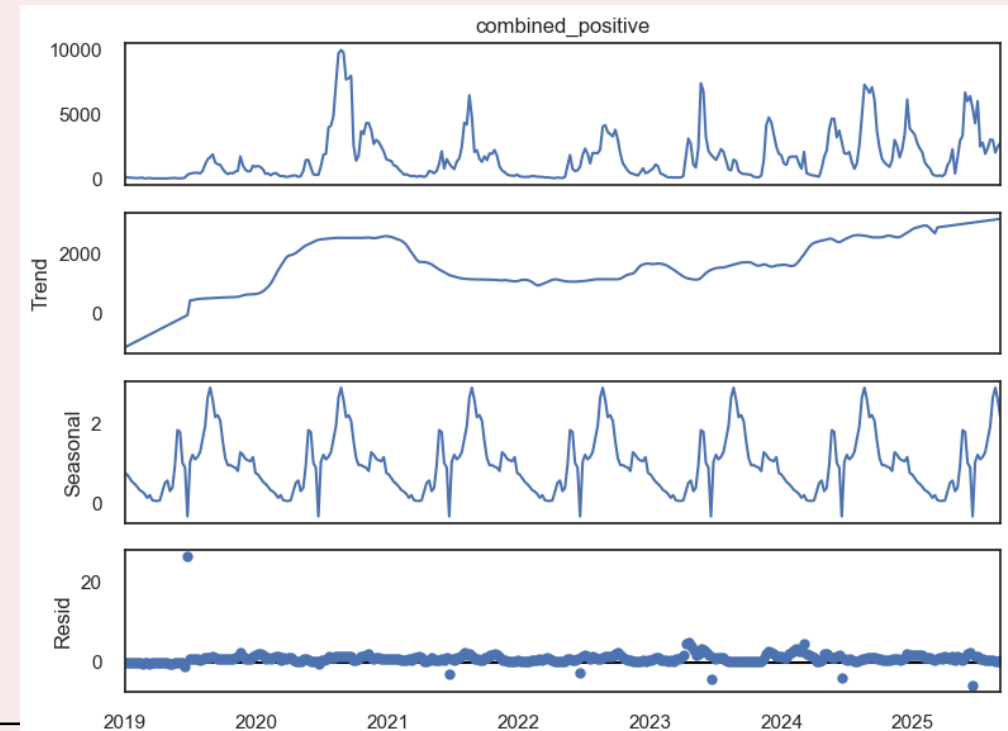
•Additive

- Clear trend and seasonality:** Malaria cases show a rising long-term trend and a stable annual seasonal cycle.
- Residuals reveal shocks:** Large spikes in residuals around 2020–2021 and 2023 suggest episodic outbreaks.
- Modeling implication:** Most systematic variation is captured by trend and seasonality



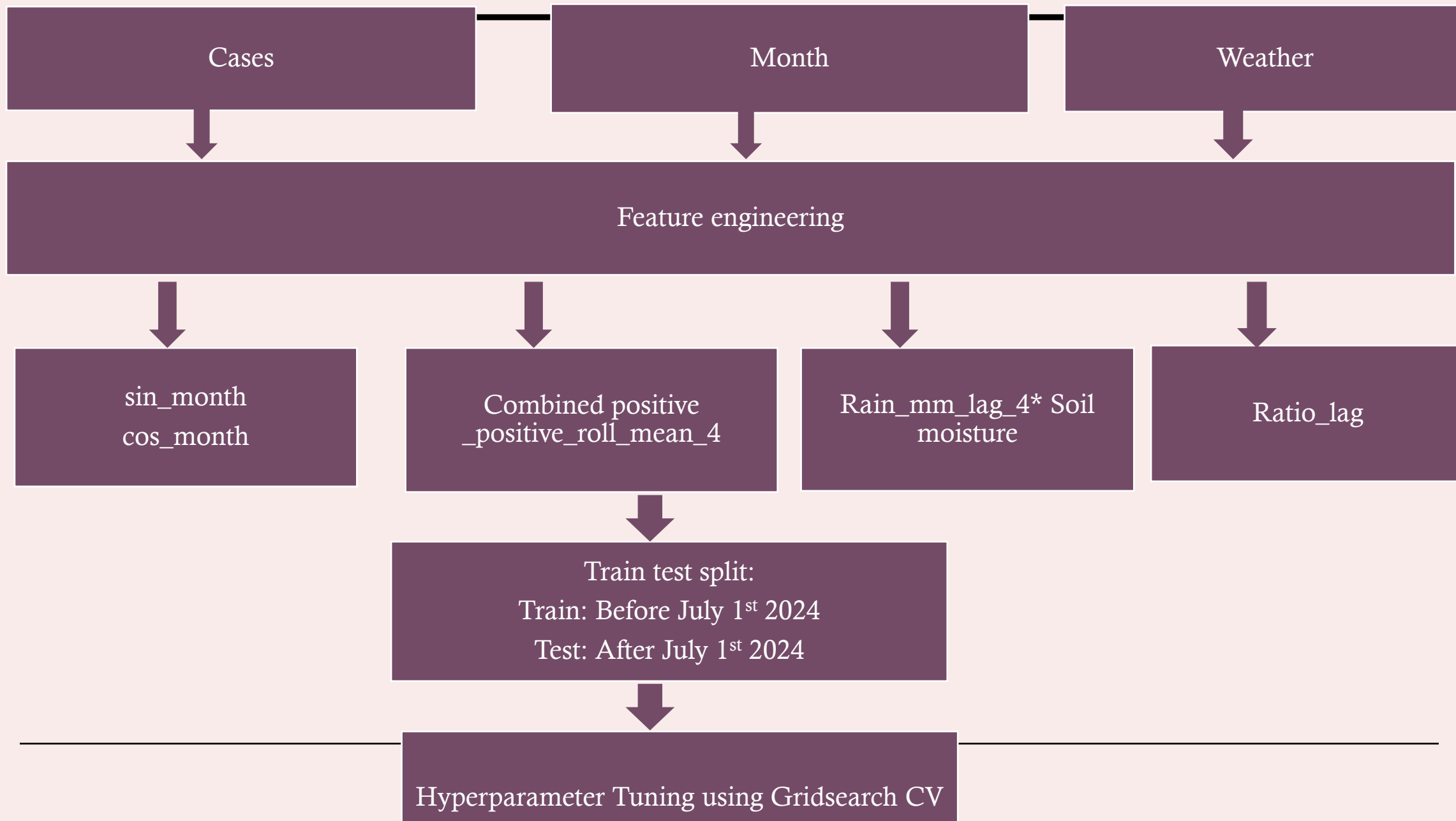
•Multiplicative

- Seasonality scales with level:** Malaria peaks are 2–3× higher than baseline during seasonal highs, indicating multiplicative seasonality.
- Trend shows long-run increase:** Rising baseline from 2020 to 2025 suggests growing transmission risk over time.
- Residuals are minimal:** Most variation is explained by trend and seasonality.



FEATURE ENGINEERING AND TUNING





MODELS

Model	R*2	MASE	Precision	Recall	ROC-AUC
Light GBM					
Validation	0.76	1.63	0.50	0.50	0.67
Test	0.71	0.93	0.33	0.50	0.55
HGB					
Validation	0.72	1.80	0.46	0.83	0.76
Test	0.65	1.06	0.32	1.00	0.71
XGBoost					
Validation	0.76	1.61	0.41	0.75	0.71
Test	0.60	1.10	0.29	0.67	0.67
Radom Forest					
Validation	0.74	1.73	0.43	0.50	0.70
Test	0.75	0.90	0.43	0.50	0.83

FINAL MODEL PERFORMANCE

Radom Forest (Final)	R*2	RMSE	MASE	F1 Score	Precision	Recall	ROC-AUC
Test (Regression)	0.99	193	0.23		0.61	0.85	0.87
Test (Classificati on)							

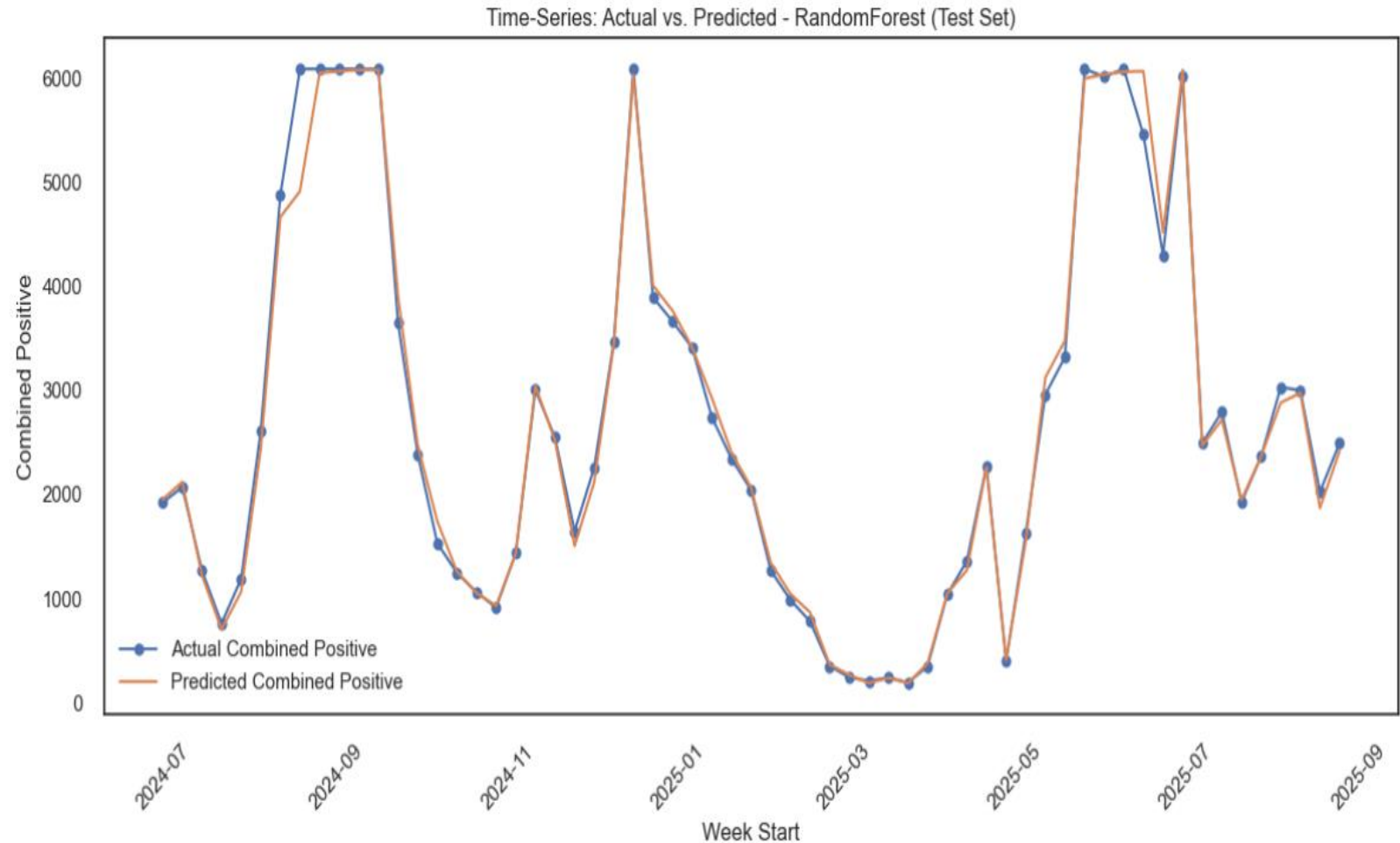
TIME-SERIES: ACTUAL VS. PREDICTED - RANDOMFOREST

- High Accuracy**

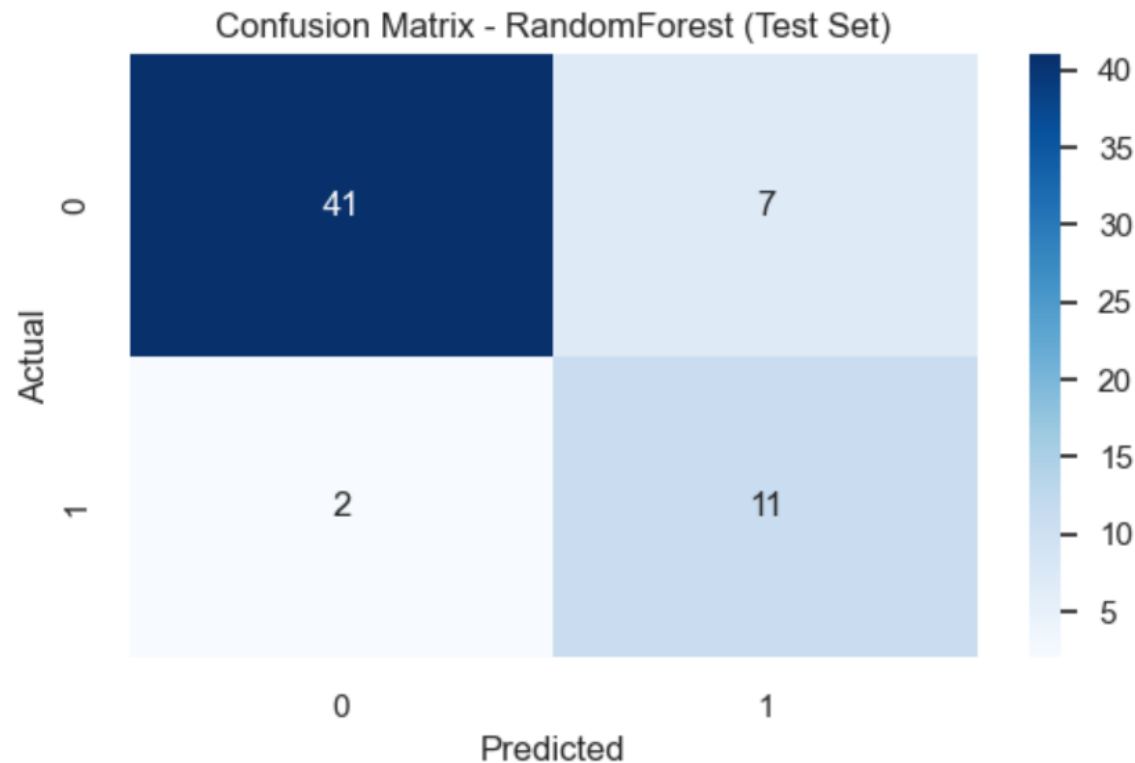
Predicted values closely match actual data, indicating strong model performance.

- Predictive Utility**

Reliable forecasts support planning for health interventions and resource allocation.



CONFUSION MATRIX



- High True Negative Rate**

Majority of negative cases correctly identified (41 out of 50).

- Low False Negative Count**

Only 2 actual positives missed, indicating good sensitivity.

- Balanced Performance**

Model shows reasonable precision and recall, suitable for deployment with minor tuning

Kakuma Ward Malaria Predictor

Predict malaria cases or detect outbreaks (next week's cases ≥ 2000 and ≥ 1.5 times current week's cases) using weekly weather data.

Number of Weeks (1-4):

4

Task:

Get the Case Counts

Current Malaria Cases:

0

Week 1

Temperature (°C):

25.1234

Relative Humidity (%):

60.1234

Rainfall (mm):

10.1234

Wind Speed (km/h):

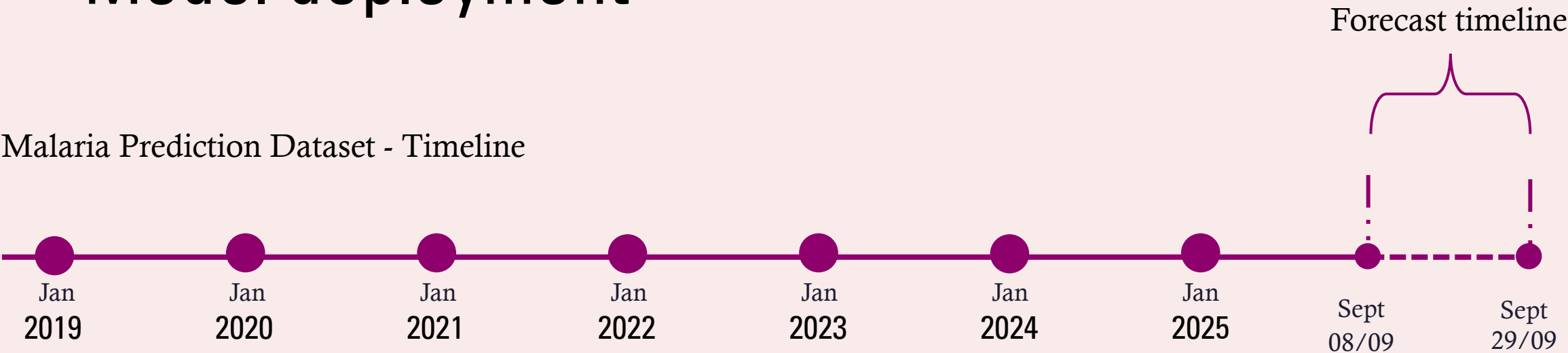
Business utility (for the health facility management):

- I. Resource allocation. This include things like drugs, test kits and nets
- II. Prepositioning of response supplies ahead of an outbreak or upsurge in cases
- III. Timing of interventions such as insecticide spraying and net distribution.
- IV. Evaluate impact of interventions in averting high morbidity.
- V. Operational logistics: Human resources, vehicles, fuel

<https://kakumamalariapredictor-e4116b68755a.herokuapp.com/>

Model deployment

Malaria Prediction Dataset - Timeline



Forecasted
Weather Data

week_start	temp_c	rh_pct	rain_mm	wind10_kmh	soil_moisture_top_m3 m3	Malaria cases
9/1/2025	28.29404762	51.83333333	5.4	7.726785714	0.086607143	2776
9/8/2025	27.9297619	54.49404762	6	9.0125	0.07864881	?
9/15/2025	29.66666667	43.96428571	0.3	9.168452381	0.066065476	?
9/22/2025	29.20357143	45.17857143	22.5	8.714880952	0.084261905	?
9/29/2025	27.32678571	58.69642857	13.7	6.888095238	0.110565476	?

RECOMMENDATIONS

1. Invest in Electronic Medical Records (EMRs)

- **Why:** Real-time malaria case counts are essential for accurate forecasting and timely alerts.
- **Recommendation:** Deploy EMRs in high-burden facilities starting with Kakuma, integrated with DHIS2 and the predictive model.
- **Impact:** Enables automated data ingestion, reduces reporting delays, and improves model responsiveness.

2. Upgrade to Deep Learning Models

- **Why:** Daily case counts allow for finer-grained predictions and anomaly detection.
 - **Recommendation:** Experiment with LSTM or Temporal Fusion Transformers for time-series forecasting.
 - **Impact:** Captures complex temporal patterns and improves early warning accuracy.
-

3. Strengthen Multiagency Collaboration

- **Why:** Weather and environmental data are critical predictors.
- **Recommendation:** Formalize data-sharing agreements with:
 - **Kenya Meteorological Department** (real-time weather)
 - **Kenya Space Agency** (satellite imagery, vegetation indices)
- **Impact:** Enhances model inputs and supports geospatial risk mapping.

4. Add Geospatial Visualization

- **Why:** Maps help localize outbreaks and guide targeted interventions.
- **Recommendation:** Integrate GIS tools to display predicted hotspots by ward or village.
- **Impact:** Improves decision-making for vector control and resource deployment.

5. Build an Alert System

- **Why:** Timely alerts enable rapid response.
 - **Recommendation:** Develop an SMS/email alert system linked to model outputs.
 - **Impact:** Notifies health managers and CHWs of predicted spikes for early action.
-

6. . Draft Operational SOPs

- **Why:** Alerts must trigger structured responses.
- **Recommendation:** Create SOPs for:
 - Resource prepositioning
 - CHW mobilization
 - Outbreak response protocols
- **Impact:** Ensures consistency and accountability in response efforts.

7. Translate Insights for CHWs

- **Why:** CHWs are frontline responders.
 - **Recommendation:** Simplify model outputs into actionable messages (e.g., “Expect spike in week 21—prepare for outreach”).
 - **Impact:** Empowers CHWs to act early and reduce morbidity.
-

THANK YOU

Malaria does not have to be deadly. Timely interventions can reduce disease burden and mortality. Predictive disease models can help us intervene better.

Collaborators:

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6. Joy Sila

